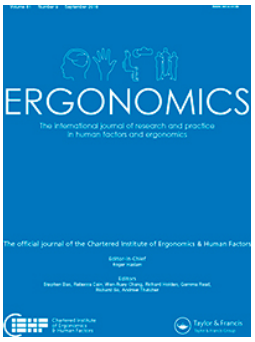

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Towards better microsleep predictions in fatigued drivers: exploring benefits of personality traits and IQ

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ABSTRACT
Fatigued driving is one of the main contributors to road traffic accidents. Poor sleep quality and lack of sleep negatively affect driving performance, and extreme states of fatigue can cause microsleep (i.e., short episodes of sleep with complete loss of awareness). Driver monitoring systems analyse biosignals (e.g., gaze, blinking, heart rate) and vehicle data (e.g., steering wheel movements, lane holding, acceleration) to detect states of fatigue and prevent accidents. We argue that inter-individual differences in personality, sensation seeking behaviour, and intelligence could improve microsleep prediction, in addition to sleepiness. We tested 144 male participants in a supervised driving track after 27 hours of sleep deprivation. More than 74% of drivers experienced microsleep, after an average driving time of 52 min. Overall, prediction models for microsleep vulnerability and driving time before microsleep were significantly improved by conscientiousness, sensation seeking and non-verbal IQ, in addition to situational sleepiness, as individual risk factors.

Practitioner summary: This study offers valuable insights for the design of driver monitoring systems. The use of individual risk factors such as conscientiousness, sensation seeking, and non-verbal IQ can increase microsleep prediction. These findings may improve monitoring systems based solely on physiological signals (e.g., blinking, heart rate) and vehicle data (e.g., steering wheel movements, acceleration, cornering).

Abbreviations: ADAC: Allgemeiner Deutscher Automobil Club; ANOVA: analysis of variance; AIC: Akaike information criteria; CI: confidence interval; GPS: global positioning system; IQ: intelligence quotient; IQR: inter quartile range; KSS: Karolinska sleepiness scale; NEO-PI-R: revised NEO personality inventory; OLS: ordinary least squares; PSQI: Pittsburgh sleep quality index; SPM: standard progressive matrices; SSS: sensation seeking scale; WHO: World Health Organization

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Fatigued driving; microsleep; personality traits; sleep deprivation; supervised driving; driver monitoring

1. Introduction

Driving is a psychologically demanding activity that requires not only fine-motor abilities but also cognitive skills and high performance in attention, visual search, and reaction time (Karthaus et al. 2020; Milosevic 1997). Driving efficiency and safety is not only influenced by physical abilities and driving experience but also by psychological states such as stress and fatigue (Havârneanu, Măirean, and Popuşoi 2019; Mollicone et al. 2019), cognitive workload (Foy and Chapman 2018; Liao et al. 2018), and even drivers' personality traits like anger, conscientiousness, extraversion, and sensation seeking (Găianu, Giosan and Sârbescu 2020; Demir, Demir, and Özkan 2016). Previous research has suggested various models linking driver personality traits to driving safety, mediated by aggressive driving or the propensity to disregard norms (Demir, Demir, and Özkan 2016). Along these lines we propose a link between drivers' personality traits and driving safety mediated by psychological states such as fatigue and sleepiness.

The global status report on road safety, highlighted road traffic injuries as the primary cause of death among people aged 5–29 years, with an average of 1.35 million annual road traffic deaths (World Health Organization 2018). Further, an awareness rising campaign from the General German Automobile Club (ADAC) and the research centre for sleep medicine at the University of Regensburg determined driver

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fatigue and sleepiness as one of the major reasons for vehicle collisions (16% of all traffic accidents with heavy goods vehicles). They furthermore estimated sleep deprivation of 17 hours to result in a decrease of driving performance comparable to a blood alcohol concentration of 0.05 g/dl, and 24 hours of sleep deprivation comparable to 0.10 g/dl (Zulley 2012; Philip and Åkerstedt 2006), way over the legal limits in the UK (0.8 g/dl) and most of Europe (0.5 g/dl).

Although the terms fatigue, drowsiness and sleepiness are not always clearly distinguished in the literature, we refer to fatigue as a state of task related exertion and drop in performance after long times on task, that can be improved by a short break with a change of activity (Brown 1994). Sleepiness on the other hand refers to the experienced need for restitutive sleep and can be influenced by the time since prior wake, duration and quality of prior sleep (or the lack of it), time of day, individual circadian type, as well as by fatigue itself (Brown 1994; Matthews et al. 2012; Zion, Drach-Zahavy, and Shochat 2018). Apart from dangerously decreasing cognitive and physical performance endangering traffic safety, sleepiness can also result in total loss of consciousness, i.e., microsleep (Craig et al. 2006; Grandjean 1979). Microsleep is a temporary episode of sleep, lasting from a fraction of a second up to a few seconds, where an individual loses awareness followed by a sudden and frightened shift to wakefulness (Durmer and Dinges 2005; Golz et al. 2011; Schleicher et al. 2008; Tirunahari et al. 2003). While clinical definitions of microsleep are based on distinct changes in brain activity, several behavioural markers related to facial expression, head movement and eyelid closure have been successfully used to detect episodes of microsleep events (Skorucak et al. 2020). During microsleep episodes, the individual is unable to process any sensory input, which can result in extremely dangerous situations when in a setting that demands constant awareness. Driver fatigue intensifies extreme states of driver sleepiness and is thus a major issue for traffic accidents (Brown 1994). The detection and prediction of fatigue and sleepiness, as well as potentially mediating personality traits and their relationship are of critical importance in traffic safety research.

An interdisciplinary research field is emerging from computer science, transportation research, ergonomics, and psychology, fostering biosignal analysis and affective computing to detect relevant patterns and psychological states in drivers (Hidalgo-Gadea 2017; Poria et al. 2017; Skorucak et al. 2020). Most state recognition approaches for driver monitoring systems analyse fatigue related parameters in behaviour and psychophysiology. Heart rate variability, eyelid opening, blinking behaviour, pupillometry, and facial expression have shown to be useful markers to detect states of fatigue (Awasekar et al. 2019; Hidalgo-Gadea 2017; McIntire et al. 2017; Michail et al. 2008; Skorucak et al. 2020; Vyas and Woodson 2010). A main shortcoming of such systems is their focus on detection of present events, rather than prediction of upcoming dangerous situations, as a risk factor.

In addition to monitoring driver's current states while waiting for dangerous situations to happen, prior knowledge about driver's stable traits may additionally provide useful insights about individual risk factors for unsafe driving and microsleep vulnerability, which in turn may be useful to adapt and fine-tune detection systems for upcoming microsleep events. Therefore, the identification of such risk factors could help develop personalised algorithms for preventive driver monitoring systems. Recently, a major car insurance company in Germany started offering bonuses for cautious driving behaviour using smartphone accelerometry data and GPS to track individual driving behaviour (ADAC 2020). Similar to their severity weighting of harsh cornering or hard breaking depending on the time of day and the road type, we believe that individual risk factors and up-to-date sleep logs could help fine-tuning monitoring systems for personalised warnings to individually dangerous situations.

To this end, in the present study we analyse a microsleep data set, collected during a real driving experiment with sleep deprived subjects. A pool of inter- and intra-individual parameters including demographic data, sleep behaviour, subjective sleepiness, personality and intelligence scores were linked to fatigued driving performance and microsleep events in a supervised driving track. While sleepiness is expected to causally trigger microsleep, the exact effect of over-night sleep deprivation time and individual sleep quality is further explored for its predictive value on microsleep vulnerability. Furthermore, inter-individual traits such as the five factor personality traits, sensation seeking behaviour, and intelligence are hypothesised to play a significant role, either as a risk factor or protective mechanism for microsleep vulnerability as well as total driving time until microsleep.

2. Methods

A group of 174 male volunteers, age between 18 and 34 years ($M = 24.15$, $SD = 3.70$), were recruited through

Figure 1. Driving track facilities shut off from the public road network. The course length was approximately 2 km long and nearly circular, driving direction was clockwise, and speed was restricted to 20 km/h.

social media for an experimental driving trial. Participants needed to have a valid driving licence and individuals with physical or mental disabilities or severe sleep-related disorders (see PSQI scores in Table A1 in Appendix A) were excluded from the study. All participants signed informed consents and were compensated with 100€ for their involvement during the approximately 24 hours of experiment. Information about all aspects and risks of the study was disclosed, excluding the researchers' intention to induce and record microsleep events. Complete data was available for 144 participants after outlier removal. See complete descriptive statistics in Table A1 in Appendix A.

Participants arrived at 6pm to the trial site at a private driving track facility in Germany (Figure 1) and were sleep deprived until the next morning, following a strict on-site sleep deprivation protocol. Participants were informed in advance about the overnight stay but were not aware of any sleep deprivation. They were asked to bring their pyjamas to conceal the purpose of the overnight stay and were instructed to follow a typical daily routine before the experiment (i.e., usual wake time, sleep duration, and naps). Consumption of caffeine and alcohol was prohibited during the entire trial. The next morning, starting between 9 and 11am, participants conducted a supervised 4-hour driving trial with a safety instructor on the passenger's seat continuously monitoring for dangerous situations. The driving trial ended after participants experienced microsleep or a maximum of 4 hours of driving was completed. For safety reasons, participants had a mandatory recovery time of 2–6 hours before leaving the facility.

High fatigue levels during the driving trial were experimentally induced by the prior sleep deprivation. Participants were given detailed instructions to stay awake and remain within the facilities overnight, while completing several online tests and questionnaires presented on a computer at prespecified times. Completion of the sleep deprivation protocol was controlled by the time stamps for each completed online questionnaire, and smartwatches controlled the absence of prolonged periods of inactivity and their GPS location. Total prior wake time was individually assessed from on-site sleep deprivation time and their reported individual wake-up time the day before. Sleep deprivation time was used to (1) induce high fatigue levels, as well as to (2) collect demographic data, and (3) conduct computer-based test batteries every 30 minutes over a period of 9 hours. Drivers' habitual smoking and alcohol consumption frequency were assessed on a 4-point Likert scale and individual driving skill was rated by the experimenter (e.g., lane holding, line crossing and risk in curves) during a familiarisation baseline drive the day before the experiment.

The Pittsburgh Sleep Quality Index (PSQI; Buysse et al. 1989) was used to assess usual sleep quality, habitual sleep efficiency and sleep disturbances during the past 4 weeks. With 19 items about sleep habits in the last month, total PSQI scores were calculated.

The Revised NEO Personality Inventory (NEO-PI-R; Costa and McCrae 1992) consisted of 240 items and was used to assess the big five personality traits (i.e., Neuroticism, Extraversion, Openness to Experience, Conscientiousness and Agreeableness).

The Sensation Seeking Scale (SSS; Zuckerman et al. 1964) consisted of 40 items arranged in 4 subscales (i.e., Thrill and Adventure Seeking, Disinhibition, Experience Seeking and Boredom Susceptibility) describing sensation seeking behaviour. It was used as a trait to describe the readiness to take risks for the sake of experiences.

The Standard Progressive Matrices (SPM; Raven, Raven, and Court 1998) were the original version of Raven's Progressive Matrices test developed to assess language independent fluid intelligence. The SPM

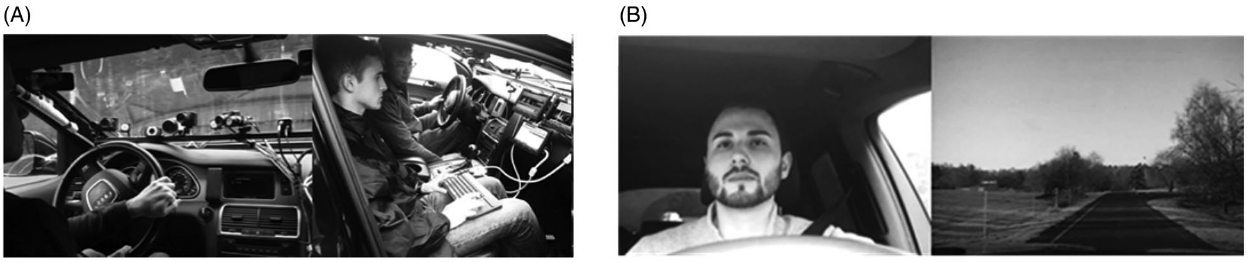


Figure 2. (A) Camera setup to monitor drivers' sleepiness and microsleep events via facial expression, blinking, and driving behaviour. The passenger's seat was also equipped with an additional break-pedal to secure the vehicle in case of microsleep. (B) Face- and road-video used to monitor drivers' fatigue and driving behaviour from monitors on the passenger's seat and in post-processing to detect microsleep events.

consisted of 60 matrices in which participants are asked to identify the missing element that completes a matrix pattern. Items were presented progressively in ascending difficulty, and total SPM scores were compared to standard table values to estimate a non-verbal fluid intelligence component of Spearman's general intelligence.

The Karolinska sleepiness scale (KSS; Åkerstedt and Gillberg 1990) was used as a measure of situational sleepiness throughout the experiment. Participants indicated on a 10-point scale, which sleepiness level best reflected their current psycho-physical state (from 1 = very alert to 10 = very sleepy, can't stay awake). KSS was assessed at the evening of arrival, the morning after sleep deprivation, right before starting the driving trial and after every 30 minutes of driving.

For the actual experiment an Audi Q7 was set up as test vehicle with several camera systems and a monitoring station at the front passenger seat (Figure 2(A)). Timestamps for microsleep events were extracted offline, from face and road video post-processing by means of behavioural observation of facial expression and unusual driving behaviour (Figure 2(B)). A microsleep classification scale based on behavioural criteria was developed (Skorucak et al. 2020; Schleicher, Galley, Briest, and Galley 2008). Prolonged eyelid closure ($>$ 1000 ms), slow drifting head movements, excessive relaxation of facial muscles, and loss of attention were behavioural cues protocolled to classify microsleep. This assessment of microsleep events from video material was performed consistently by the first author and cross-checked by the third author for 20% of the cases. Only one ambiguous case was identified with respect to the exact starting time of the event, not the occurrence itself.

Participants who experienced microsleep while driving were labelled as microsleep vulnerable and their driving time until microsleep occurred was recorded. Drivers not experiencing microsleep within the 4-hours trial were labelled microsleep resistant (after the same sleep deprivation time). A safety instructor on the passenger's seat continuously monitored the driver's fatigue level and was instructed to secure the vehicle in case of microsleep or dangerous driving behaviour.

Data analysis was performed in R version 3.6.3 (R Core Team 2019). Script and data are provided upon request and all libraries used are listed in Supplementary Appendix B. Differences in KSS Sleepiness across time were analysed with robust ANOVAs (Wilcox 2017). Hierarchical logistic regression models were computed to investigate the unique contribution of each parameter on microsleep vulnerability and hierarchical linear regressions were performed to assess their unique contribution to the total driving time before experiencing microsleep. Final prediction models were computed with parameters which previous' unique contribution was significant at an $\alpha = 0.10$ level and further robust models are reported in Appendix A as confirmatory results (Wilcox 2017). Forest plots are provided for each regression model to visualise odds ratios and standardised coefficients, respectively, with their corresponding 95% confidence intervals. This attempt to complement classical statistical significance with visual information on the point estimate of the effect size should help the reader avoid misleading categorical inferences (see Amrhein, Greenland, and McShane 2019).

3. Results

Table A1 in Appendix A provides an overview of sample demographics. For instance, 35.41% of the participants reported being bad sleepers (PSQI $>$ 5), while only 4.17% reported excessive sleep problems (PSQI $>$ 10; see Hinz et al. 2017). The sample mean in Neuroticism corresponded to the 76th percentile in a

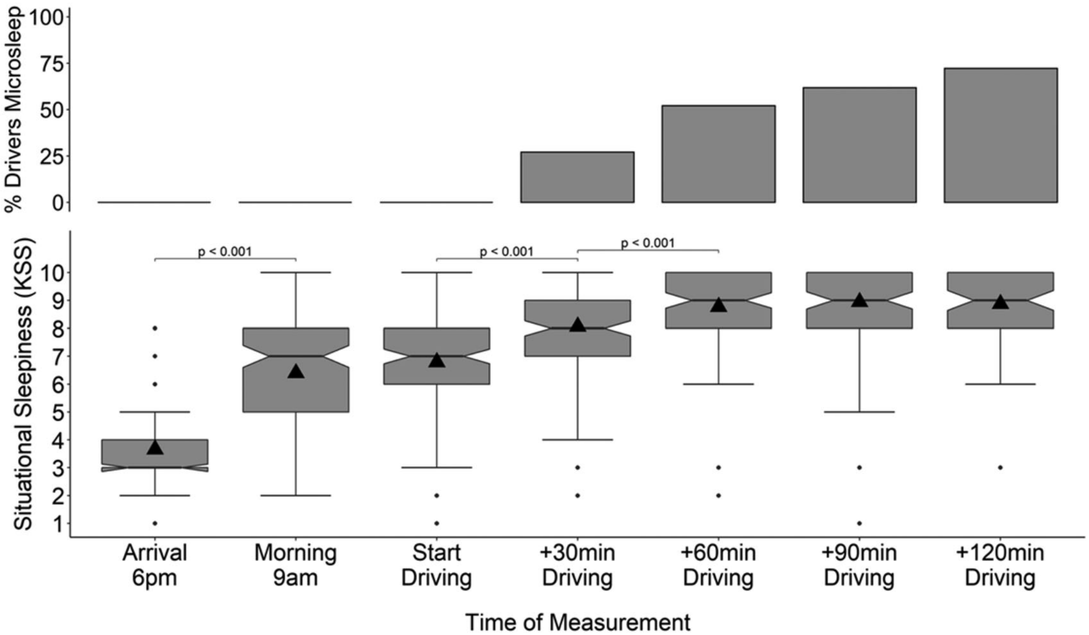


Figure 3. Karolinska Sleepiness Scores (below) and cumulative number of drivers fallen asleep (above) across time. Pairwise comparisons were estimated by robust *post hoc* tests (see Mair and Wilcox 2020), black triangles indicate the group mean and notches represent a 95% confidence interval for medians ($1.58 \times IQR/\sqrt{n}$).

normative sample for adult men, and was therefore slightly higher than expected. Sample mean in Extraversion corresponded to an average 38th percentile, and the sample means of Openness to Experience (18th percentile), Conscientiousness (14th percentile) and Agreeableness (24th percentile) were lower than the normative average. Further analyses were conducted with raw scores rather than normative standard values. The sample mean $M = 23.57$ ($SD = 5.22$) corresponded to a slightly high sensation seeking score (SS > 23). The sample IQ score in non-verbal intelligence was $M = 103.08$, $SD = 16.87$.

Up to the start of the driving trial, participants were sleep deprived for $M = $ 27:43 h ($SD = $ 2:57 h). A repeated measures robust ANOVA for 15% trimmed means conducted with 2000 bootstraps (see Mair and Wilcox 2020) revealed significant differences in sleepiness across time, $F = 163.72$, $F_{crit} = 2.48$, $p < .05$. Further robust post hoc tests revealed significantly increased KSS scores after sleep deprivation ($M = 6.40$, $SD = 1.87$) relative to the night before ($M = 3.66$, $SD = 1.36$), $\hat{\psi} = -2.85$, 95%CI $[-3.85, -1.85]$, $p < .001$ (Figure 3). Furthermore, drivers' sleepiness increased significantly during the progression of the driving trial within the first 30 minutes, $\hat{\psi} = -1.34$, 95%CI $[-1.91, -0.77]$, $p < .001$, and between 30 and 60 min, $\hat{\psi} = -0.77$, 95%CI $[-1.21, -0.32]$, $p < .001$. Note a ceiling effect in KSS after 60 min driving time and the increasing number of drivers experiencing microsleep and reducing the sample size (i.e., 52% after 60 min, 61% after 90 min and 72% after 120 min).

To control for the variability in the time of start driving (9–11 am), we confirmed that delayed start driving increased the individual total prior wake time, $r(144) = .47$, $p < .001$. However, a multiple regression revealed that start driving time had no significant effect on KSS at start driving after controlling for the mediating effect of total prior wake time, $F(1,141) = 1.46$, $p = .229$. The individual start driving time had also no direct effect on time to microsleep after controlling for total prior wake time, $F(1,104) = 2.41$, $p = .123$.

During the course of the experiment, a total of 107 drivers (74.31%) experienced at least one episode of microsleep while driving, and 37 microsleep resistant drivers (25.69%) completed the 4-hour trial without any sign of dangerous driving behaviour. Microsleep vulnerable drivers experienced their first microsleep event after $M = $ 51:31 min average driving time ($SD = $ 36:34 min, $Med = $ 41:24 min, $IQR = $ 49:12 min), see Figure 4. The occurrence of microsleep events was dichotomised rather than counted, because the first microsleep event was often dangerous enough to finish the trial and stop driving, as should happen in real driving conditions.

A closer look at participants' sleepiness across time reveals a significant main effect of over-night sleep deprivation, $F(1,\ 40.27) = 159.55$, $p < .001$, but no

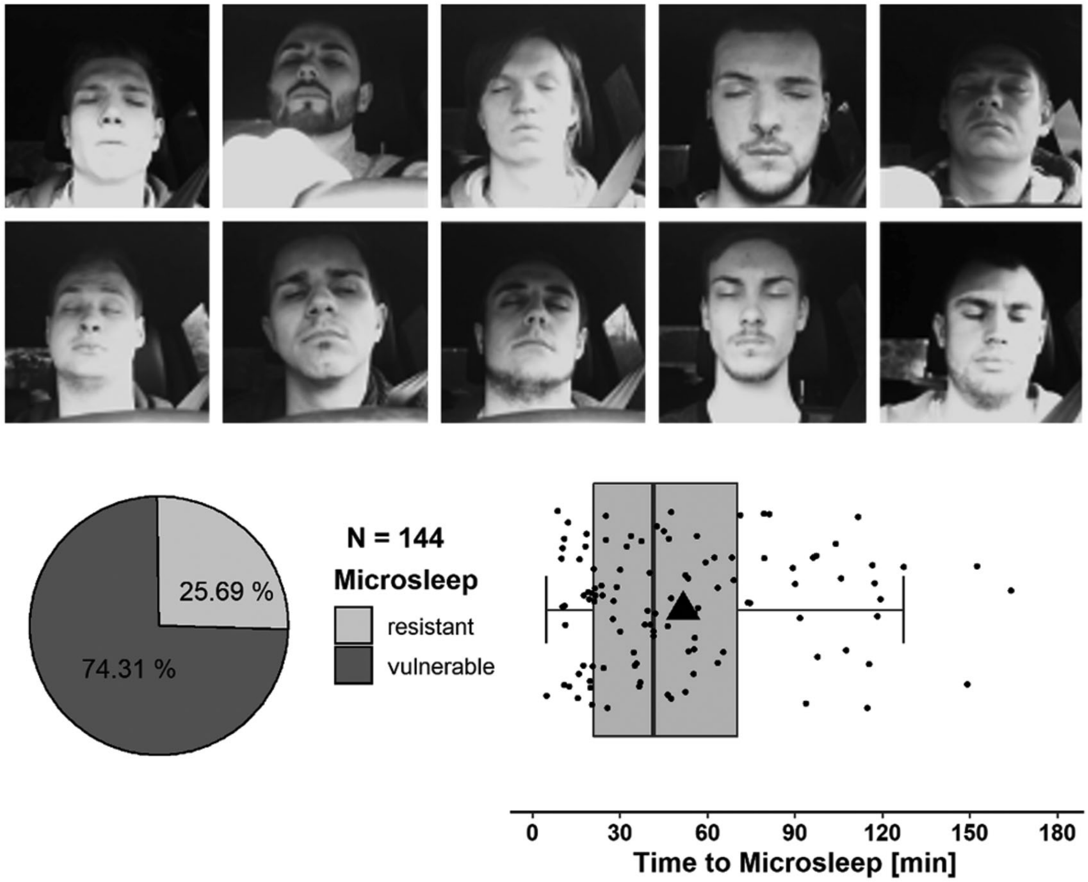


Figure 4. Drivers' facial expressions during microsleep events, incidences of microsleep vulnerable drivers and distribution of total driving time before first microsleep event, black triangles indicate the group mean.

significant interaction between microsleep vulnerability group and sleep deprivation in a robust two-way ANOVA with 15% trimmed means, $F(1, 40.27) = 0.01$, $p = .916$. More interestingly, a further robust two-way ANOVA between morning KSS and KSS at start driving showed a significant interaction, $F(1, 47.44) = 4.98$, $p = .031$ and no main effect, indicating that microsleep vulnerable drivers maintained their high sleepiness level, whereas microsleep resistant drivers see their KSS scores decreased at the time driving starts, as shown in Figure 5. Thus, the relevant predictive value for microsleep vulnerability is not sleepiness after sleep deprivation itself, but rather the relative decrease in sleepiness at start driving. During the course of the experiment, sleepiness increases progressively in both groups, with microsleep resistant drivers maintaining their advantage of the initial reduction in sleepiness over time.

The first modelling step of the hierarchical multiple logistic regression was performed to investigate which demographic characteristics, if any, could predict microsleep vulnerability and microsleep resistance in fatigued drivers. The overall model was not significantly better than the baseline without predictors, $\chi^2(8) = 7.36$, $p = .499$, $\Delta AIC = +8.70$, and neither the drivers' age, height, weight, driving skills, sleep quality (PSQI), sleep deprivation time, nor smoking or drinking behaviour had a significant unique contribution to predicting microsleep vulnerability. These demographic variables were nevertheless maintained as a control set in further modelling steps to investigate the unique contribution of later variables.

The addition of morning sleepiness scores to the demographic model in the second step was no significant improvement in prediction performance, $\chi^2(1) = 0.03$, $p = .858$, $\Delta AIC = +1.90$, but a third regression step including both, morning KSS and sleepiness at start driving, resulted in an overall decrease of AIC and a significant improvement of the loglikelihood ratio, $\chi^2(1) = 4.71$, $p = .030$, $\Delta AIC = -2.70$, see Table 1. Note that, conceptually, in this last sleepiness model, the unique effect of KSS start driving, from which morning KSS is outpartialized, additionally contributes the difference in sleepiness between both KSS measures at start driving to the model, as shown in Figure 5.

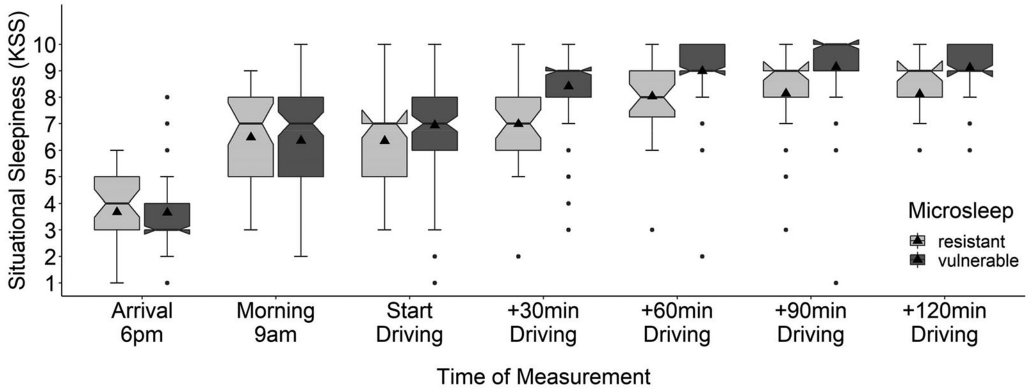
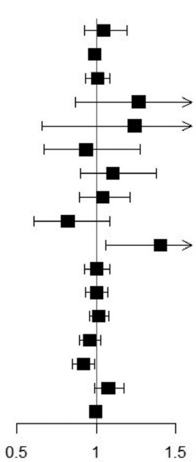


Figure 5. Self-reported situational sleepiness across time and between groups, black triangles indicate the group mean and notches represent a 95% confidence interval for medians.

Table 1. Hierarchical multiple logistic regression of microsleep vulnerability.

Model	Model test	p	AIC	ΔAIC	R^2 McFadden	R^2 Cox and Snell	R^2 Nagelkerke
1: Demographics	$\chi^2(8)=7.36$	.499	178.40	+8.70	.04	.05	.07
2: Morning KSS	$\chi^2(1)=0.03$	.858	180.70	+1.90	.05	.05	.07
3: Start driving KSS	$\chi^2(1)=4.71$	.030	176.00	−2.70	.07	.08	.12
4: Full model	$\chi^2(7)=10.1$	.180	179.80	+3.80	.14	.14	.21

Model Parameter	b	se	95% CI b	Odds	z	p	95% CI odds ratio
4: Full model							
Intercept	5.18	11.83	[−17.99, 28.82]		0.44	.661	
Age	0.05	0.06	[−0.07, 0.18]	1.05	0.76	.449	
Weight	−0.01	0.02	[−0.04, 0.03]	0.99	−0.57	.572	
Height	0.01	0.04	[−0.07, 0.08]	1.01	0.20	.839	
Smoking	0.24	0.20	[−0.14, 0.65]	1.27	1.18	.238	
Alcohol	0.22	0.32	[−0.42, 0.86]	1.24	0.67	.501	
Driving skills	−0.06	0.16	[−0.39, 0.25]	0.94	−0.40	.688	
PSQI	0.10	0.11	[−0.10, 0.32]	1.10	0.92	.358	
Sleep deprivation time	0.04	0.08	[−0.11, 0.19]	1.04	0.53	.596	
Morning KSS	−0.20	0.15	[−0.50, 0.08]	0.82	−1.35	.176	
Start driving KSS	0.34	0.15	[0.06, 0.64]	1.41	2.31	.021	
Neuroticism	0.01	0.04	[−0.07, 0.08]	1.00	0.06	.955	
Extraversion	0.01	0.04	[−0.07, 0.07]	1.00	0.04	.968	
Openness to experience	0.02	0.03	[−0.04, 0.08]	1.02	0.55	.586	
Agreeableness	−0.04	0.04	[−0.11, 0.03]	0.96	−1.17	.243	
Conscientiousness	−0.08	0.04	[−0.16, −0.01]	0.92	−2.10	.036	
Sensation seeking	0.07	0.04	[−0.01, 0.16]	1.08	1.71	.087	
Non-verbal IQ	−0.01	0.01	[−0.03, 0.02]	1.00	−0.16	.876	



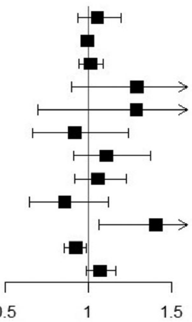
Note. Predictors in steps 1–3 are omitted for better display. Model improvement was tested from differences in log-likelihood ratios between each step and the previous.

To identify further possible risk factors inhibiting drivers to reduce sleepiness on start driving (or rather protective factors helping to mobilise resources), non-verbal IQ (SPM), the big five personality traits (NEO-PI-R) and Sensation Seeking Behaviour (SSS) were included in a last modelling step. The full logistic regression model was not a significant improvement in prediction compared to the previous model, $\chi^2(7) = 10.16$, p=.180, $\Delta AIC = +3.80$, but revealed several predictors of interest with considerable unique contribution, see Table 1.

A final regression model was computed with selected predictors of the previous models with considerable unique contributions (Table 1). Eventually, predictive performance of the final model was favoured over statistical significance of single predictors, and a more liberal significance level of $\alpha = 0.10$ was used to select useful factors. The final model included Conscientiousness and Sensation Seeking Behaviour as inter-individual factors and was significantly better in predicting microsleep vulnerability than a model considering drivers' sleepiness (step 3; $\chi^2(2) = 8.04$, p = .018,

Table 2. Final logistic regression model of microsleep vulnerability.

	b	se	95% CI b	Odds	z	p	95% CI odds ratio
Final model							
Intercept	0.74	7.15	[−13.46, 14.86]		0.10	.918	
Age	0.05	0.06	[−0.07, 0.18]	1.05	0.83	.405	
Weight	−0.01	0.02	[−0.04, 0.03]	0.99	−0.51	.609	
Height	0.01	0.04	[−0.06, 0.09]	1.01	0.38	.704	
Smoking	0.26	0.19	[−0.11, 0.65]	1.29	1.34	.182	
Alcohol	0.25	0.31	[−0.36, 0.88]	1.29	0.81	.419	
Driving skills	−0.09	0.16	[−0.41, 0.21]	0.92	−0.56	.577	
PSQI	0.10	0.10	[−0.09, 0.32]	1.11	0.99	.321	
Sleep deprivation time	0.06	0.07	[−0.09, 0.20]	1.06	0.75	.451	
Morning KSS	−0.16	0.14	[−0.44, 0.11]	0.86	−1.11	.268	
Start driving KSS	0.34	0.15	[0.06, 0.64]	1.41	2.35	.019	
Conscientiousness	−0.08	0.04	[−0.16, −0.01]	0.92	−2.17	.030	
Sensation seeking	0.07	0.04	[−0.01, 0.15]	1.07	1.63	.104	



Note. AIC = 179.80, $R^2_{McFadden} = 0.12$, $R^2_{Cox\ and\ Snell} = 0.13$, $R^2_{Nagelkerke} = 0.19$.

$\Delta AIC = -4.00$) and demographics only (step 1; $\chi^2(4) = 12.78$, $p = .012$, $\Delta AIC = -4.80$). See Table 2 for the individual regression coefficients.

Results of the final logistic model with maximum likelihood estimator were validated by a robust logistic regression with a weighted sample and a Mlqe estimator). Table A2 in Appendix A provides an overview of all robust regression coefficients and respective 95% CIs and odds ratios, confirming the significant contribution of conscientiousness in predicting microsleep vulnerability.

For those 107 microsleep vulnerable participants, driving time until first microsleep varied greatly between 00:04:48 and 02:43:48 hours with an interquartile range of 49 minutes. Following the same hierarchical approach mentioned above, multiple linear regression models were computed to predict total driving time from demographics, sleepiness, and inter-individual personality traits.

Neither demographics (step 1; $F(8,98) = 0.95$, $p = .482$, $\Delta AIC = +8.03$), nor morning sleepiness (step 2; $F(1,97) = 0.04$, $p = .849$, $\Delta \mathrm{AIC} = +1.96$), were a significant improvement in predictive performance of time to microsleep. And, like microsleep vulnerability, time to microsleep was significantly decreased by the difference between morning sleepiness and sleepiness at start driving, thus significantly improving the overall prediction $F(1,96) = 8.88$, $p = .004$, $\Delta AIC = -7.46$.

The full model (step 4) extended by non-verbal IQ (SPM), the big five personality traits (NEO-PI-R) and Sensation Seeking Behaviour (SSS) was no direct improvement in predictive performance per se, $F(7,89) = 1.75$, $p = .108$, $\Delta AIC = +0.20$, but revealed interesting predictors with considerable unique contributions, see Table 3. For instance, higher driving skills ($\beta = 0.19$, $p = .060$) and higher IQ ($\beta = 0.19$, $p = .058$) seemed to prolong driving time and thus delayed microsleep, whereas Conscientiousness and Sleepiness at start driving significantly reduced driving time.

A final linear regression model was computed with selected predictors of the previous models with considerable unique contributions (Table 4). Predictive performance of the final model was again favoured over statistical significance of single predictors, and a significance level of $\alpha = 0.10$ was used to select useful factors. This final model included Conscientiousness and IQ as inter-individual factors and significantly improved prediction performance of time to microsleep over a model considering sleepiness (step 2; $F(2,94) = 5.17$, $p = .007$, $\Delta AIC = -7.17$), *or* demographics only (step 1; $F(4,94) = 5.01$, $p = .001$, $\Delta AIC = -12.67$, see Table 4).

Residuals of the final model are more or less normally distributed with few outliers skewing the distribution (see Figure 6). Additionally, the errors show a heteroscedastic pattern with increasing residual variance for increasing time predictions. To validate the previous results of the OLS model, a robust regression was computed with improved Theil-Sen estimator with back-fitting, the Harrell-Davis intercept estimation, and a percentile bootstrap method with $n = 2000$ to calculate 95% CIs (see Wilcox 2017). Table A3 in Appendix A provides an overview of robust regression coefficients and respective 95% CIs.

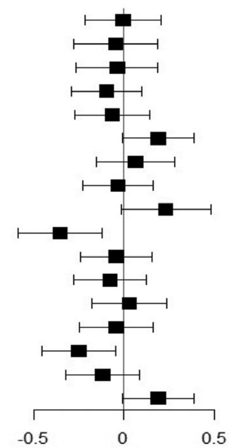
4. Discussion

With the present study we wanted to explore the possibility of using personality data to enhance the prediction of microsleep events during driving. Unlike simulator studies, sleep-deprived participants drove a real car in a closed-circuit, safe driving environment, enabling us to

Table 3. Hierarchical multiple linear regression of time to microsleep.

Model	Model test	*p*	AIC	ΔAIC	R^2	$R^2_{adjusted}$
1: Demographics	F(8,98)=0.95	.482	1084.91	+8.03	.07	.00
2: Morning KSS	F(1,97)=0.04	.849	1086.87	+1.96	.07	.00
3: Start driving KSS	F(1,96)=8.88	.004	1079.41	−7.46	.15	.06

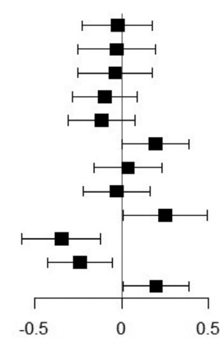
Model parameter	*b*	*se*	95% CI b	*β*	*t*	*p*	95% CI β
4: Full model							
Intercept	301.47	195.97	[−87.92, 690.9]		−1.54	.128	
Age	−0.05	1.04	[−2.10, 2.01]	−0.01	−0.04	.965	
Weight	−0.10	0.28	[−0.65, 0.45]	−0.04	−0.37	.711	
Height	−0.20	0.60	[−1.40, 1.00]	−0.04	−0.33	.745	
Smoking	−2.87	3.00	[−8.82, 3.09]	−0.09	−0.96	.341	
Alcohol	−3.13	5.08	[−13.22, 6.96]	−0.06	−0.62	.539	
Driving skills	4.60	2.41	[−0.19, 9.39]	0.19	1.91	.060	
PSQI	0.98	1.66	[−2.31, 4.27]	0.06	0.59	.557	
Sleep deprivation time	−0.39	1.19	[−2.76, 1.98]	−0.03	−0.33	.745	
Morning KSS	4.39	2.36	[−0.29, 9.07]	0.23	1.86	.066	
Start driving KSS	−6.94	2.31	[−11.52, −2.36]	−0.35	−3.01	.003	
Neuroticism	−0.25	0.59	[−1.43, 0.94]	−0.04	−0.41	.680	
Extraversion	−0.47	0.62	[−1.70, 0.75]	−0.08	−0.77	.443	
Openness to experience	0.13	0.47	[−0.80, 1.06]	0.03	0.28	.783	
Agreeableness	−0.24	0.60	[−1.43, 0.94]	−0.04	−0.41	.684	
Conscientiousness	−1.45	0.60	[−2.64, −0.26]	−0.25	−2.43	.017	
Sensation seeking	−0.82	0.72	[−2.24, 0.60]	−0.12	−1.14	.256	
Non-verbal IQ	0.42	0.21	[−0.02, 0.85]	0.19	1.92	.058	



Note. Predictors in steps 1–3 are omitted for better display. Model improvement was tested from differences in sum of squares between each step and the previous.

Table 4. Final linear regression model of time to microsleep.

	b	*se*	95% CI b	*β*	*t*	*p*	95% CI β
Final model							
Intercept	195.94	122.22	[−46.72, 438.6]		1.60	.112	
Age	−0.23	1.00	[−2.21, 1.76]	−0.02	−0.23	.822	
Weight	−0.06	0.26	[−0.59, 0.46]	−0.03	−0.24	.810	
Height	−0.20	0.58	[−1.35, 0.95]	−0.04	−0.34	.734	
Smoking	−2.90	2.87	[−8.60, 2.79]	−0.10	−1.01	.314	
Alcohol	−5.52	4.70	[−14.85, 3.82]	−0.11	−1.17	.244	
Driving skills	4.66	2.33	[−0.04, 9.29]	0.19	2.00	.048	
PSQI	0.57	1.52	[−2.43, 3.58]	0.04	0.38	.705	
Sleep deprivation time	−0.33	1.17	[−2.65, 2.00]	−0.03	−0.28	.782	
Morning KSS	4.75	2.29	[0.20, 9.30]	0.25	2.07	.041	
Start driving KSS	−6.77	2.24	[−11.22, −2.33]	−0.34	−3.03	.003	
Conscientiousness	−1.39	0.55	[−2.59, −0.30]	−0.24	−2.52	.013	
Non-verbal IQ	0.43	0.21	[0.02, 0.84]	0.20	2.10	.039	



Note: AIC = 1072.24, R^2=0.23, R^2_{adjusted}=0.14.

study real microsleep events in an ecologically valid situation. More than 74% of drivers experienced microsleep after an average driving time of 52 min. We successfully identified interindividual risk and/or protective factors that improve prediction of the occurrence of microsleep events (i.e., conscientiousness and sensation seeking behaviour) as well as the total driving time before microsleep (i.e., conscientiousness and non-verbal IQ). These factors are discussed in more detail below.

The implemented (non-supervised) overnight sleep deprivation protocol succeeded in sleepiness induction. Participants not only reported significantly higher sleepiness on the day of the experiment compared to the night before (Figure 3), but the protocol also successfully induced a large number of microsleep events during the experiment (in 107 drivers, 74% of the sample). Furthermore, our results satisfactorily display a continuous increase in drivers' sleepiness over time and, subsequently, a progressive increase in the number of microsleep events over time-on-task. These findings are in line with similar studies evaluating fatigued driving in track settings (Kecklund and Åkerstedt 1993; Sagaspe et al. 2008; Shekari Soleimanloo et al. 2019). Overall, study design and experimental setup effectively allowed collecting valuable data on microsleep events in fatigued drivers in a safe but real driving environment, with implemented safety measures successfully enabling participants to

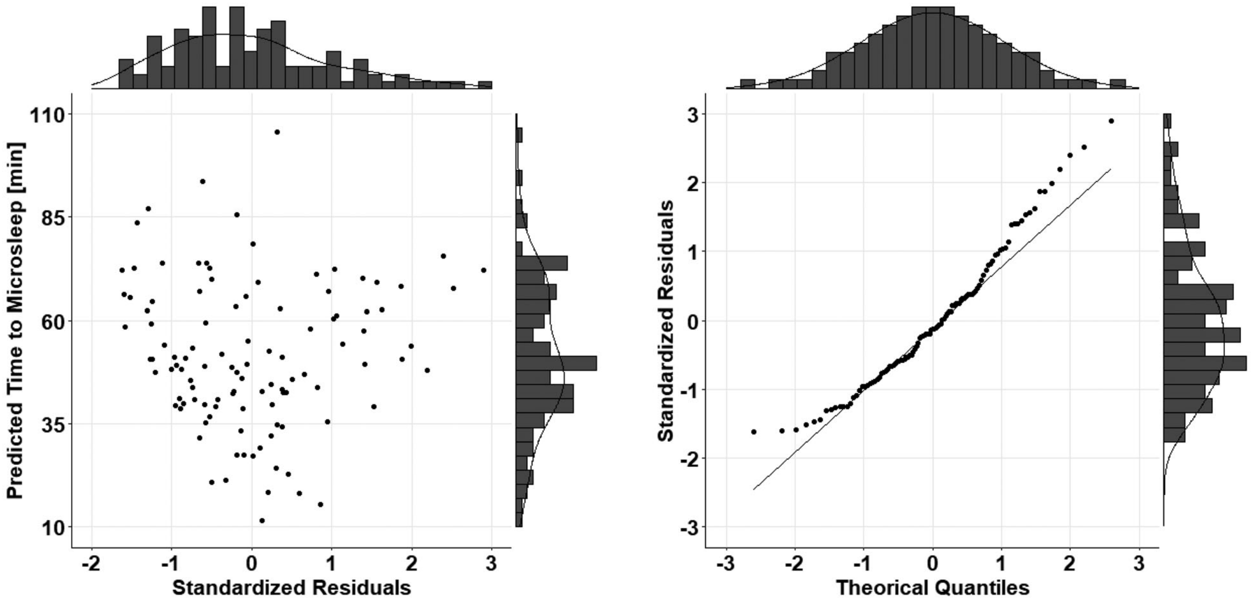


Figure 6. Scatter plots of standardised residuals for the final linear model. Heteroscedasticity and deviations from normality were controlled with robust regression.

experience an ecologically valid setting compared to driving simulators.

Regarding factors influencing microsleep vulnerability and time to microsleep, demographic indicators do not seem to be very promising predictors. Neither age, nor height or weight had notable effects on either outcome variable, but note the limitations in sample demographics. Other demographic factors, such as individual driving skills may prolong time to microsleep (although the effect was not confirmed in robust model), but do not seem to be a useful predictor for overall microsleep vulnerability. This makes sense, assuming that drivers with higher skills and experience may be better able to detect signs of emerging microsleep and, therefore are better able to compensate and strategically activate resources countering effects of sleepiness to some extent (Hamid et al. 2016).

Habitual alcohol use and smoking habits also did not contribute significantly in our final models. However, taking into consideration the odds ratios and confidence intervals in the forest plots (Tables 2 and 4), frequent alcohol consumption and smoking behaviour tend to increase the likelihood of microsleep events and similarly shorten time to microsleep. Rather than prematurely concluding that these non-significant results show no effect of alcohol and smoking behaviour, we encourage the reader to carefully interpret the compatibility and the direction of these effects at the given level of uncertainty (Amrhein, Greenland, and McShane 2019). While alcohol consumption was prohibited during the experiment for obvious reasons, smoking was not. Future research should build on these results while meticulously controlling for drinking and smoking behaviour, as well as experienced craving and urges to consume, which may in turn cause restlessness (He et al. 2019; Purani, Friedrichsen, and Allen 2019), counteracting sleepiness. Similarly, the individual effect of caffeine withdrawal was not controlled for and may be interesting to consider in future work.

Surprisingly, regular sleep quality (PSQI) also failed to show any effect. This result was unexpected, given the direct influence of sleep quality on daytime sleepiness (Dewald et al. 2010). However, the influence of PSQI on daytime sleepiness – and thereby on microsleep vulnerability – is likely to have been surpassed by the substantial sleep deprivation experimentally induced in the present sample. More importantly, it is noteworthy that neither the duration of sleep deprivation, nor the absolute morning sleepiness proved to be useful predictors of microsleep in our models. While the failure of finding effects for duration of sleep deprivation might be explained by the lack of variance in this factor due to our experimental design (all participants experienced roughly the same amount of sleep deprivation), the fact that experienced morning sleepiness was not a good predictor seems counterintuitive. However, combined with the finding that the relative decrease between morning sleepiness and sleepiness at start driving time presents a more sensible picture. Those resistant to microsleep seem to be able to lower their experienced sleepiness once they start driving, see Figure 5. This crucial decrease in sleepiness is statistically independent of any other parameter in our regression models, hence not explainable by any of the personality variables we tested. This effect may be attributed to the mediating link of attitudes and subjective norms such as danger concerns and normative compliance, allowing to (temporarily) mobilise necessary resources (Demir, Demir, and Özkan 2016; Găianu, Giosan and Sârbescu 2020). Alternatively, fluctuations in sleepiness between morning sleepiness and sleepiness at start driving may be explained by individual circadian phases, which could be independent from any of the personality variables tested.

Situational sleepiness was specifically assessed twice, once at 9 am for morning sleepiness and again at start driving, to compensate for the variability in start driving time between individuals during the experiment. Drivers starting earlier may have had a higher sleep propensity than drivers starting mid-morning, while the latter would have had a longer sleep deprivation (up to the exact start driving time). We checked this possibility by showing a non-significant effect of start driving time on sleepiness at start driving, after controlling for total prior wake time. The variability in start driving time also had no direct effect on time to microsleep. Although the effect of time of day on sleep propensity and drivers' sleepiness may have been levelled out by the induced sleep deprivation in our study, effects of individual circadian rhythms cannot be ruled out completely (see limitation below).

Turning to personality traits, we targeted the BIG-5 personality traits, sensation seeking, and non-verbal intelligence in the present study. Our results suggest that conscientiousness can be regarded as an inter-individual protective factor against microsleep vulnerability. This is in harmony with the idea that people high in conscientiousness, for example, follow socially prescribed norms for impulse control and are able to delay gratification (Roberts et al. 2009). With respect to driving when fatigued, this could translate into being better able to activate resources to stay on task compared to individuals that are less conscientious. This is also in line with findings indicating that habitual patterns of behaviour (i.e., personality and intelligence) are connected to volition and related processes (e.g., Baumann and Kuhl 2002; Maples-Keller et al. 2016).

The effect that conscientiousness is also related to reduced time to microsleep seems counterintuitive at first. However, we argue that for the subset of individuals high in conscientiousness their activated resources are depleted quickly, as they draw on them more heavily to maintain wakefulness compared to less conscientious individuals. One might speculate that this subgroup of conscientious people would not drive when fatigued, but only did so in an experimental setting where they wanted to perform as best as they could, very much in line with their personality. A different explanation may be a suppressor effect of subjective norms, not accounted for during the study.

Apart from conscientiousness, sensation seeking remained in our final model to predict microsleep vulnerability. Representing a higher readiness to take risks for the sake of experiences (Zuckerman 2009), it was identified as a useful and sensible indicator, although its effect on the final model was neglectable (and was not confirmed in robust model). Given that the sample in the current study consisted of mostly young man, one could argue that this homogenous group typically shows rather high values in sensation seeking (see link between sensation seeking and testosterone in young men in Campbell et al. 2010) and that a higher variability in a more diverse sample could show a more pronounced effect.

Lastly, non-verbal intelligence was associated with significantly prolonged driving time before experiencing microsleep (robust analysis confirmed marginally significant and small effect). A possible underlying mechanism could be a higher level of automatisation in driving-related behaviours, reducing the need to activate resources. However, it is important to keep in mind that the effect of non-verbal IQ on driving time was independent from individual driving skills.

Overall, personality factors seem to have a small effect but nevertheless allow to significantly improve models to predict microsleep vulnerability as well as time to microsleep. However, there are some limitations of the current study. Due to the sleep deprivation protocol and the high levels of fatigue, our collected microsleep data may not be representatively distributed, thus overrepresenting microsleep events in a normal population. On the other hand, the data reflect the dominant danger of microsleep susceptibility in fatigued drivers. In addition, participants were all young males between 18 and 34 years of age, with low demographic variability, and although driving skills were assessed, driving habits and driving experience were not controlled for during this study. Similarly, caffeine consumption was restricted during the experiment to avoid compromising self-reported sleepiness ratings and ease inducing fatigue. However, a detailed control of caffeine withdrawal symptoms was missing and may have been interesting to consider as potential risk factor together with alcohol and smoking habits.

The Pittsburgh Sleep Quality Index (PSQI; Buysse et al. 1989) was used to assess prior sleep duration, naps, and habitual sleep efficiency and sleep disturbances during the past 4 weeks prior to the experiment, but self-reports were not verified with sleep diaries, actigraphy, or polysomnography. Therefore, PSQI scores alone cannot completely rule out the possibility of mild confounding sleep disorders in the present sample. The individual circadian rhythm was not included in the analyses. In hindsight it might have been useful to clarify individual differences in the fluctuation between morning sleepiness and sleepiness at start driving. Although participants conducted a 30 minutes familiarisation trial the evening prior sleep deprivation, the analysis lacked a comparable baseline. This would have been necessary to investigate the effect of the overnight sleep deprivation on driver fatigue and microsleep, but not to investigate the role of individual differences as risk factors for microsleep. Finally, a note of caution should be sounded to avoid misconceptions about the term 'microsleep resistance' used to describe some drivers in this study. We assume that every driver is vulnerable to microsleep under extreme states of fatigue. In our study, we set thresholds at 27 hours of sleep deprivation and 4 hours of driving. However, as shown by the variability in time to microsleep, changing any of these thresholds would have influenced the incidence of microsleep and 'microsleep resistance'.

5. Conclusion

Taken together, we showed that prediction performance of microsleep events can be slightly but significantly increased utilising personality traits, in addition to the driver's situational sleepiness state. Specifically, individual factors such as driver's conscientiousness, sensation seeking behaviour, and non-verbal intelligence could help fine-tune and re-design microsleep prediction in driver monitoring systems. In our view, understanding the underlying mechanism resulting in the reduction of experienced sleepiness at the start of driving is a promising starting point for further investigation. The statistical analysis of microsleep occurrence provided in this study may be useful for the evaluation of driver monitoring systems using physiological data rather than self-reported situational sleepiness. We assume that drivers' personality traits will slightly improve predictions in driver monitoring systems too, helping interpret fatigue related physiological signals, thereby facilitating the prediction of microsleep from fatigue detection.

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Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability statement

Data and analysis script are available upon request.

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Appendix A

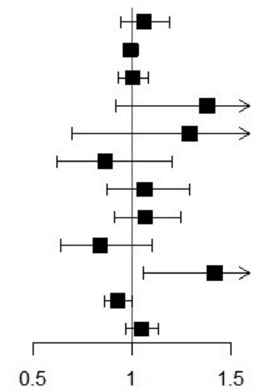
Table A1. Descriptive statistics for sample and group demographics and personality traits.

	Total sample $N = 144$		Microsleep vulnerable $n = 107$		Microsleep resistant $n = 37$	
	M	SD	*M*	SD	*M*	SD
Demographics						
Age [years]	24.06	3.73	24.05	3.72	24.08	3.82
Weight [kg]	83.38	15.18	83.38	15.51	83.35	14.38
Height [cm]	182.37	6.97	182.50	6.89	181.97	7.29
Driving Skills [1–10]*	8.00	1.25	8.00	2.00	8.00	1.00
Smoking [1–4]*	1.00	2.00	1.00	2.00	1.00	1.00
Alcohol [1–4]*	3.00	1.00	3.00	1.00	2.00	1.00
PSQI	5.14	2.27	5.29	2.39	4.70	1.87
Sleep deprivation [hours]	27.72	2.95	27.79	3.03	27.52	2.75
Personality traits						
Neuroticism	95.94	5.95	96.07	6.15	95.57	5.41
Extraversion	97.87	6.31	98.05	5.99	97.35	7.23
Openness to experience	94.81	7.61	95.24	8.09	93.54	5.93
Agreeableness	100.75	6.47	100.36	6.23	101.89	7.10
Conscientiousness	98.35	6.10	97.75	6.26	100.08	5.30
Sensation seeking	23.57	5.22	24.13	5.24	21.95	4.85
IQ	103.08	16.87	102.69	16.70	104.22	17.56

Note. Mean and standard deviation calculated for continuous variables, while the median and the inter quartile range is used to describe ordinal variables(*).

Table A2. Robust logistic regression model of microsleep vulnerability.

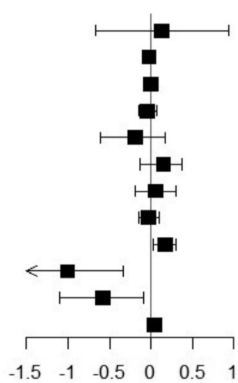
	b	*se*	95% CI b	Odds	*z*	*p*	95% CI odds ratio
Final model							
Intercept	3.06	7.36	[−11.36, 17.49]		0.42	.677	
Age	0.06	0.06	[−0.06, 0.18]	1.06	0.89	.372	
Weight	−0.01	0.02	[−0.05, 0.03]	0.99	−0.53	.598	
Height	0.01	0.04	[−0.08, 0.08]	1.00	0.07	.945	
Smoking	0.32	0.21	[−0.09, 0.74]	1.38	1.58	.114	
Alcohol	0.25	0.32	[−0.37, 0.88]	1.29	0.79	.428	
Driving skills	−0.15	0.17	[−0.48, 0.19]	0.86	−0.89	.371	
PSQI	0.06	0.10	[−0.14, 0.26]	1.06	0.60	.551	
Sleep deprivation time	0.06	0.08	[−0.09, 0.22]	1.07	0.86	.391	
Morning KSS	−0.18	0.14	[−0.45, 0.10]	0.84	−1.23	.220	
Start driving KSS	0.35	0.15	[0.06, 0.64]	1.42	2.33	.020	
Conscientiousness	−0.08	0.04	[−0.15, 0.01]	0.93	−1.98	.048	
Sensation seeking	0.05	0.04	[−0.03, 0.13]	1.05	1.10	.272	



Note. Robust coefficients computed with Mqle estimator in 10 iterations. Out of $n = 144$, 124 weights were ~ 1 and of the remaining 20 had $Med = 0.80$ and $IQR = 0.29$.

Table A3. Robust linear regression model of time to microsleep.

	b	*se*	95% CI *b*	β	*p*	95% CI β
Final model						
Intercept	158.76	94.6	[−13.36, 361.56]		.068	
Age	0.31	0.97	[−1.56, 2.20]	0.13	.705	
Weight	−0.13	0.18	[−0.47, 0.26]	−0.03	.522	
Height	0.02	0.42	[−0.88, 0.76]	0.01	.959	
Smoking	−2.00	2.59	[−6.81, 3.48]	−0.04	.471	
Alcohol	−4.57	4.66	[−14.74, 4.23]	−0.19	.291	
Driving skills	2.28	1.94	[−2.02, 5.72]	0.15	.302	
PSQI	0.70	1.54	[−2.32, 3.72]	0.06	.645	
Sleep deprivation time	−0.48	1.14	[−2.65, 1.83]	−0.03	.658	
Morning KSS	3.37	1.34	[0.65, 5.92]	0.17	.015	
Start driving KSS	−5.89	1.93	[−9.47, −1.94]	−1.01	.002	
Conscientiousness	−1.28	0.56	[−2.41, −0.19]	−0.58	.027	
Non-verbal IQ	0.38	0.22	[−0.02, 0.86]	0.04	.061	



Note. Robust coefficients computed with improved Theil-Sen estimator by replacing the sample median with the Harrell-Davis estimator and a percentile bootstrap method with $n = 2000$ to calculate 95% CIs (Wilcox 2017).